

The Changeover Playbook

How Line 4 cut changeover time by 41% in eight weeks, and how any line can repeat it.

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Introduction

A **changeover** is everything that happens between the last good part of one product and the first good part of the next. On a busy line it is easy to treat it as fixed overhead — something the shift absorbs. On Line 4 it was not small: eleven changeovers a week at an average of 94 minutes cost us 17 hours of capacity, close to two full shifts.

This playbook describes how the Line 4 team cut that time by 41% in eight weeks without buying new equipment. Most of the gain came from doing work *before* the line stopped, not from doing it faster while it was stopped. The method is simple enough to learn in an afternoon and repeatable on any line that changes product more than once a day.

The guide is written for the people who will run the next roll-outs: line leads, process engineers and the operators who own the standard. Each chapter ends with what to do next, so it can be used as a checklist as well as an explanation.

What you'll learn:

- How to see a changeover as a list of separate tasks
- How to tell work that needs the line stopped from work that doesn't
- How to move, simplify and standardise each task
- How to measure the result so the gain sticks

Who this is for:

- Line leads planning a changeover improvement
- Engineers supporting a roll-out to a new line
- Managers deciding where to invest next

Two ways to read it

If you are about to start on a line, read *The method* closely and use the table on page 5 as your worksheet. If you are deciding whether to fund a roll-out, start with *Results and roll-out*; the evidence and costs are on pages 7 and 8.

What you'll get out of it: by the end you should be able to film, split and rebuild one changeover with your own team. Expect the first one to take about two weeks from filming to a written standard.

Let's get started.

Chapter 1

The method

The method

What is a changeover?

A changeover is not one job but a chain of small ones: fetching tools, finding the next die, unclamping, lifting, cleaning, clamping, adjusting, running test parts and checking them. Each has an owner, a duration and a reason to exist. Until those tasks are written down, nobody can improve them, because everyone remembers a different version.

The single most useful question is: **does this task need the line stopped?** Work that needs the line stopped is *internal*. Work that could be done while the line is still running is *external*. On Line 4, almost half of the stopped time turned out to be external work done at the wrong moment.

Four steps

1. **Film.** Record three real changeovers end to end, from last good part to first good part. Use a phone on a tripod; nobody needs to perform for the camera.
2. **Split.** Watch the films with the operators and list every task with its duration. Mark each one internal or external.
3. **Shift.** Move every external task before the stop or after the restart. Then shorten what remains internal: quick clamps, preset tools, fewer adjustments.
4. **Standardise.** Write the new sequence as a one-page standard with photos, and train every shift on it.

Tip: start with the changeover operators complain about most, not the longest one. Early goodwill matters more than early minutes.

Internal or external?

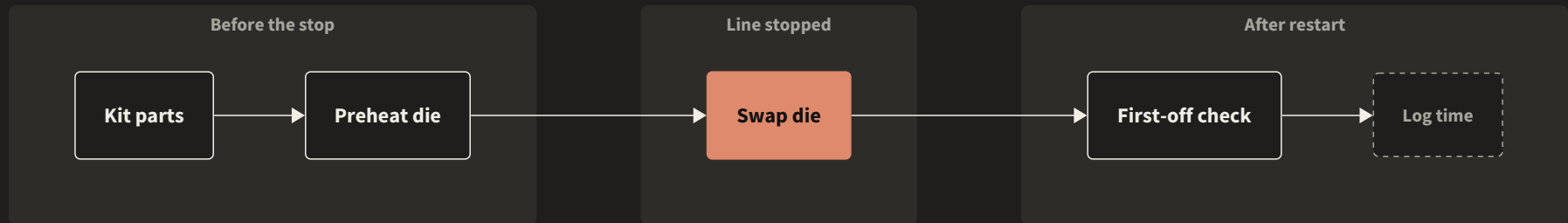
Most tasks can be decided in seconds. The table shows how the Line 4 team classified the most common ones, and what they did with each.

Task	Before	After
Fetch next die	Internal	External: kit cart
Preheat die	Internal	External: preheat station
Unclamp and swap	Internal	Internal: quick clamps
Find torque wrench	Internal	Removed: tools on cart
First-off check	Internal	Internal: gauge on cart
Log the time	Internal	External: after restart

Why kitting matters most

The biggest single gain came from a kit cart: one cart per product, loaded during the previous run with the die, clamps, tools and first-off gauge. Searching and walking disappeared from the stopped time almost entirely. It cost less than any other change the team made.

Moving work outside the stop



The rebuilt changeover: two tasks before the stop, one while stopped, two after the restart. Only "Swap die" still costs production time.

What the operators changed

The team ran the new sequence for two weeks before writing it down, adjusting it daily. Three changes made the difference:

- **Preheating off the line.** A small preheat station next to the press means the die arrives hot. This alone removed 14 minutes of waiting.
- **Quick clamps.** Replacing bolted clamps with quarter-turn clamps cut unclamping and clamping from 11 minutes to 3.
- **Gauge on the cart.** The first-off check used to wait for a gauge from the quality lab. Now the gauge travels on the kit cart with the die.

Keeping the gain

Changeover times tend to drift back up once attention moves elsewhere. Line 4 keeps the gain with three simple habits: the time of every changeover is logged on the cart's card, the line lead reviews the week's times on Monday, and any changeover over 60 minutes gets a five-minute "what slowed us down" talk at the next shift start.

***Note:** log times by hand for the first month. Operators trust a number they wrote down themselves.*

Results and roll-out

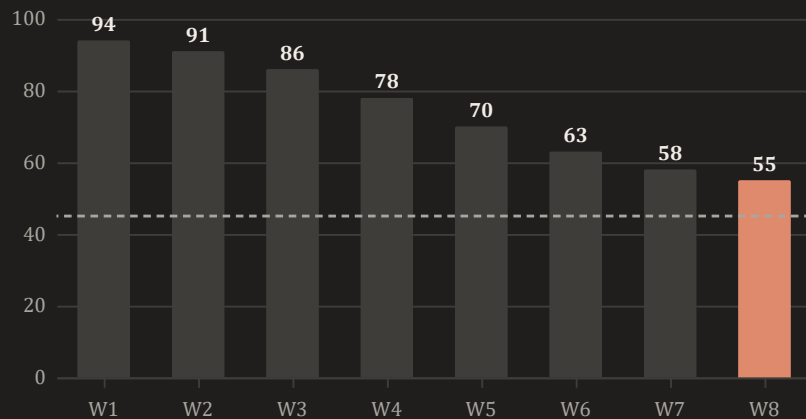
Results so far

Line 4

Average changeover time fell from 94 to 55 minutes over eight weeks, a 41% reduction. The fall was steady rather than sudden: each week the team removed one or two more sources of delay, and the new standard locked each gain in before the next change.

Average changeover time, minutes

Line 4, weekly; dashed line is the 45-minute target



At eleven changeovers a week, the 39 minutes saved per changeover return about 7 hours of production time every week — close to one full shift. The 45-minute target has not been reached yet; the remaining gap is almost entirely the die swap itself, which needs a different clamping system.

Other lines

Lines 2 and 3 started the method four weeks after Line 4 and are following the same curve more slowly. Line 5 has not improved: its changeovers are dominated by a cleaning step that the method does not address, because it is genuinely internal work.

Line	Before	Now	Change
Line 2	88 min	71 min	–19%
Line 3	102 min	84 min	–18%
Line 4	94 min	55 min	–41%
Line 5	79 min	81 min	+3%

What we learned

- **Filming is the step people skip, and the one that matters.** Every line that skipped it found fewer external tasks.
- **Operators find the best ideas.** Most of Line 4's changes came from the people doing the work.
- **A written standard is what makes it last.** Without it, times drifted back within two weeks on Line 2.
- **Not every line fits the method.** Where the slow step is truly internal, the answer is engineering, not sequencing.

Rolling it out

The plan for Q3

The proposal is to run the method on Lines 2, 3 and 5 in turn, one line at a time, with the Line 4 team coaching. Each line gets two weeks to film and split, two weeks to shift and trial, and one week to write and train the standard.

- **July:** finish Lines 2 and 3; buy six kit carts
- **August:** Line 5, focusing on the cleaning step
- **September:** audit all four lines against their standards

What it cost

The full set of changes on Line 4 cost about €6,500: one kit cart per product family (€1,100 each), quick clamps for the press, and a used preheat station. The recovered capacity paid for it within the first month.

What we need

- Approval of €40,000 for kit carts, quick clamps and two preheat stations
- A named changeover lead on each line, with two hours a week for the roll-out
- Time on the Monday production meeting for the weekly changeover review

Decision needed: approve the Q3 roll-out and the €40,000 budget by 15 June so carts can be ordered in time.

Risks

The main risk is attention: the method works when a line lead owns it and fades when nobody does. The second is Line 5, where the cleaning step may need an engineering fix that is not yet costed.

Resources

- *Line 4 changeover standard*, rev. 3 — the one-page standard with photos
- *Changeover worksheet* — the internal/external table as a printable form
- *Filming guide* — how to record a changeover in one page
- Shigeo Shingo, *A Revolution in Manufacturing: The SMED System* (1985)

Questions about this playbook go to the Operations Excellence team.